Conversation State and Memory

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Recall and choose your starting point

> Record your starting explanation for Conversation State and Memory before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

User Need and AI Contribution

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Model Memory vs Retrieval

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Hallucination Failure Types

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Conversation State and Memory trace and record the first step you can explain without help.

Use W2: Trace the worked case for Conversation **state** and retention. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Conversation State and Memory explanation against the supplied facts before discussing it.

Use the worked case for Conversation **state** and retention. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**. **State** the active values at each stage.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Conversation State and Memory target to a trace and a justified conclusion.

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Conversation State and Memory case and show the deciding evidence.

State stores topic=plants and reading_level=beginner. User requests advanced explanation.

What changes?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A learner corrects their project topic from oceans to forests. The next reply still uses oceans. What **state** should change, and what should not be saved unnecessarily?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Conversation State and Memory adds to Verified Study Buddy and keep the evidence.

Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Conversation State and Memory facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Conversation State and Memory facts and keep your reasoning.

Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Conversation State and Memory facts and keep your reasoning.

A report omits this qualification: Remembered **state**ments are not automatically current or verified; retention should follow the **state**d purpose and policy.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Conversation State and Memory facts and keep your reasoning.

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**. **State** the active values at each stage.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Conversation State and Memory facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Conversation State and Memory

> Keep a reading conversation consistent when the learner changes the selected book. Explain

> the labelled step before checking the changed condition.

Keep a reading conversation consistent when the learner changes the selected book.

In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/20

Conversation State and Memory

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First choice

Store Cedar as this session's selected book.



2. Correct choice

Replace Cedar with Maple; do not append an equally active second choice.



3. End session

Clear selected_book. A new session begins with no selected book.

CHANGE ONE THING

The next session asks "summarise my book" without naming one.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep a reading conversation consistent when the learner changes the selected book.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Clear selected_book. A new session begins with no selected book.

1. First choice

2. Correct choice

3. End session

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: End session Clear selected_book. A new session begins with no selected book. Read the picture in order. Clear selected_book.

A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The next session asks “summarise my book” without naming one. Which conclusion is supported by the supplied facts?

1. Ask which book the learner wants.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ask which book the learner

wants. The declared policy clears **session state**, so yesterday's choice is unavailable.

Repair a missing building block

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

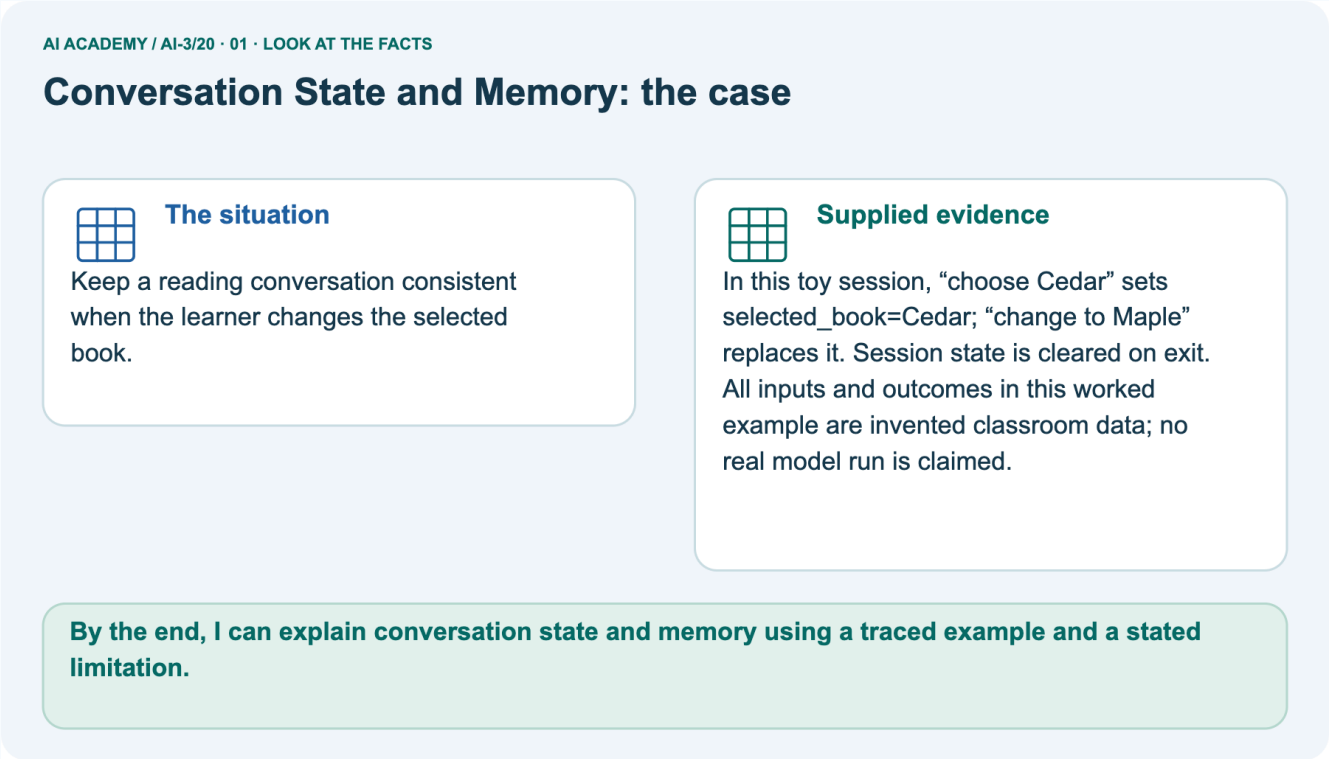
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Conversation State and Memory picture from its labels, then record a relevant limit.

01. Conversation State and Memory: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

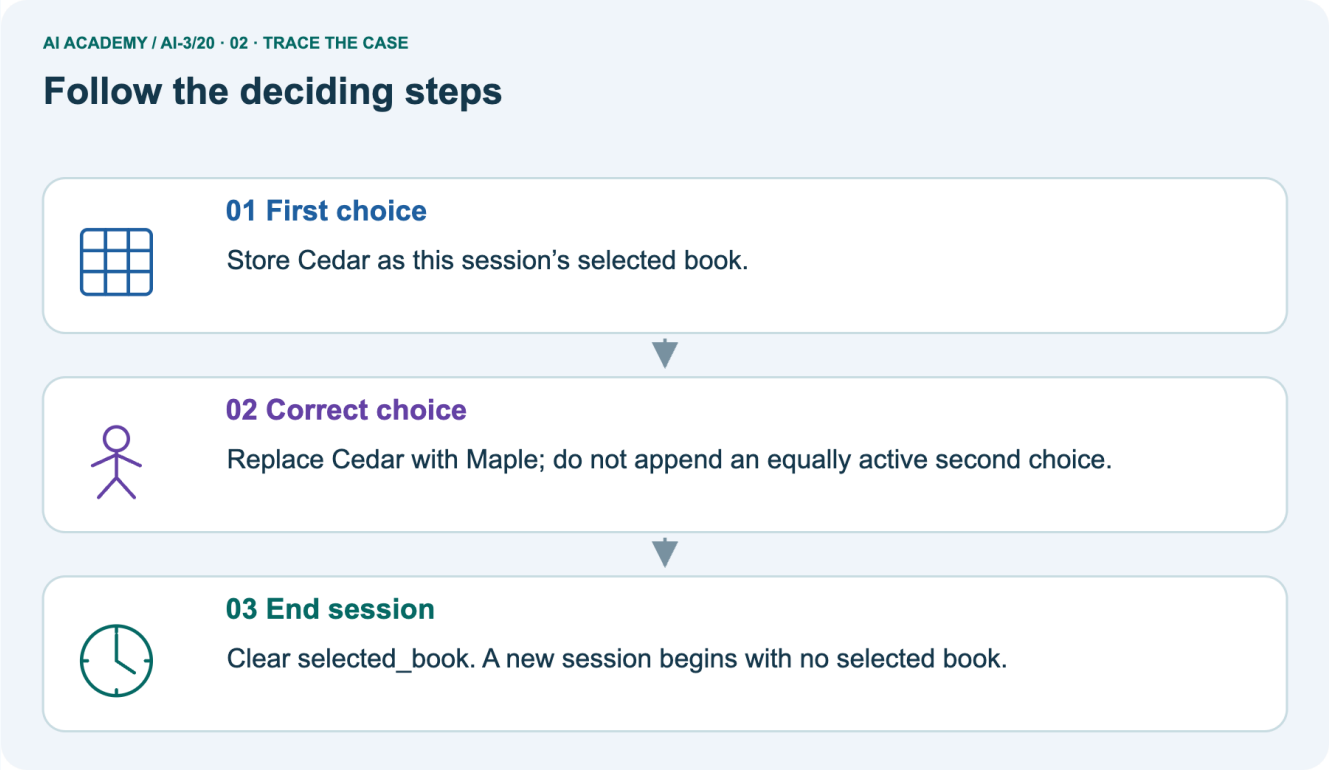
Keep a reading conversation consistent when the learner changes the selected book. Supplied facts: In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

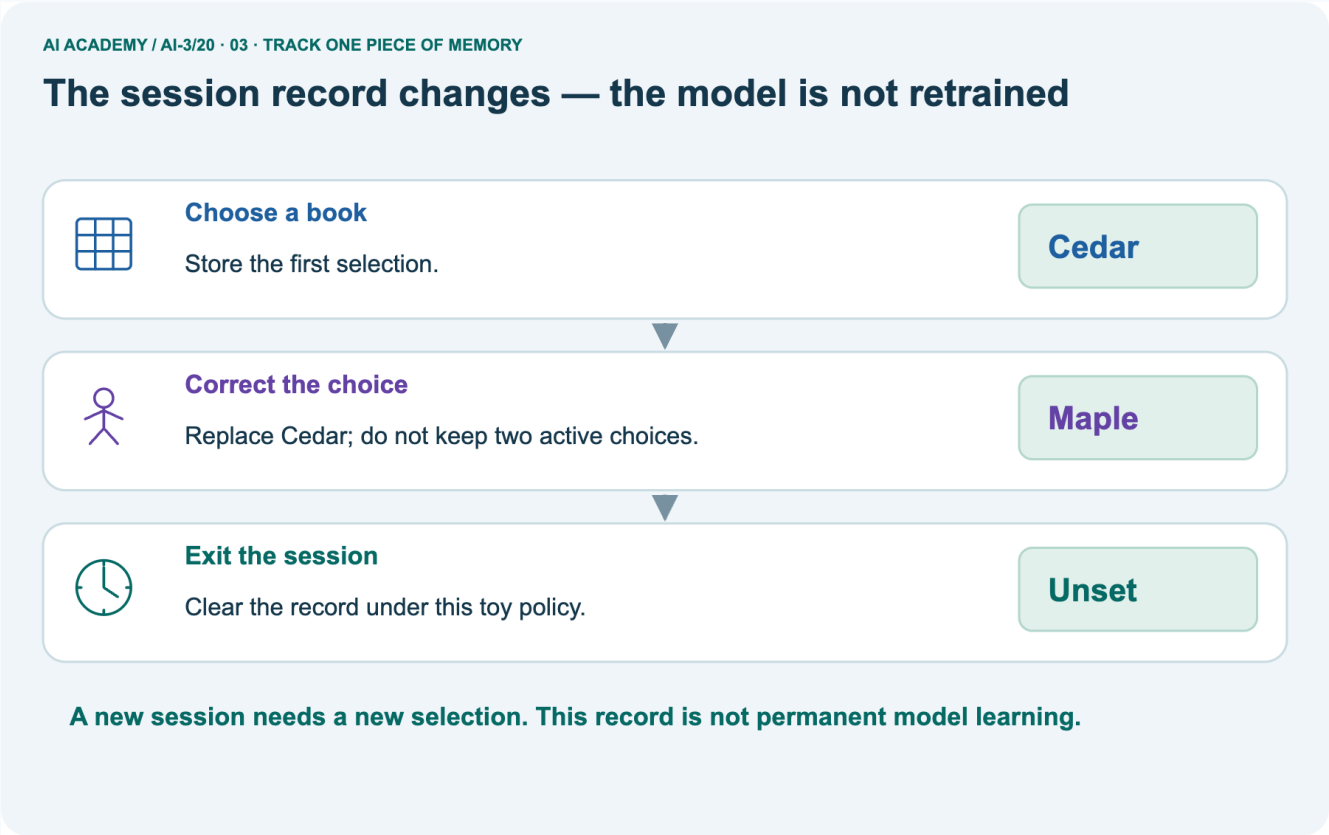
First choice: Store Cedar as this session's selected book. → Correct choice: Replace Cedar with Maple; do not append an equally active second choice. → End session: Clear selected_book.

A new session begins with no selected book.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The session record changes — the model is not retrained



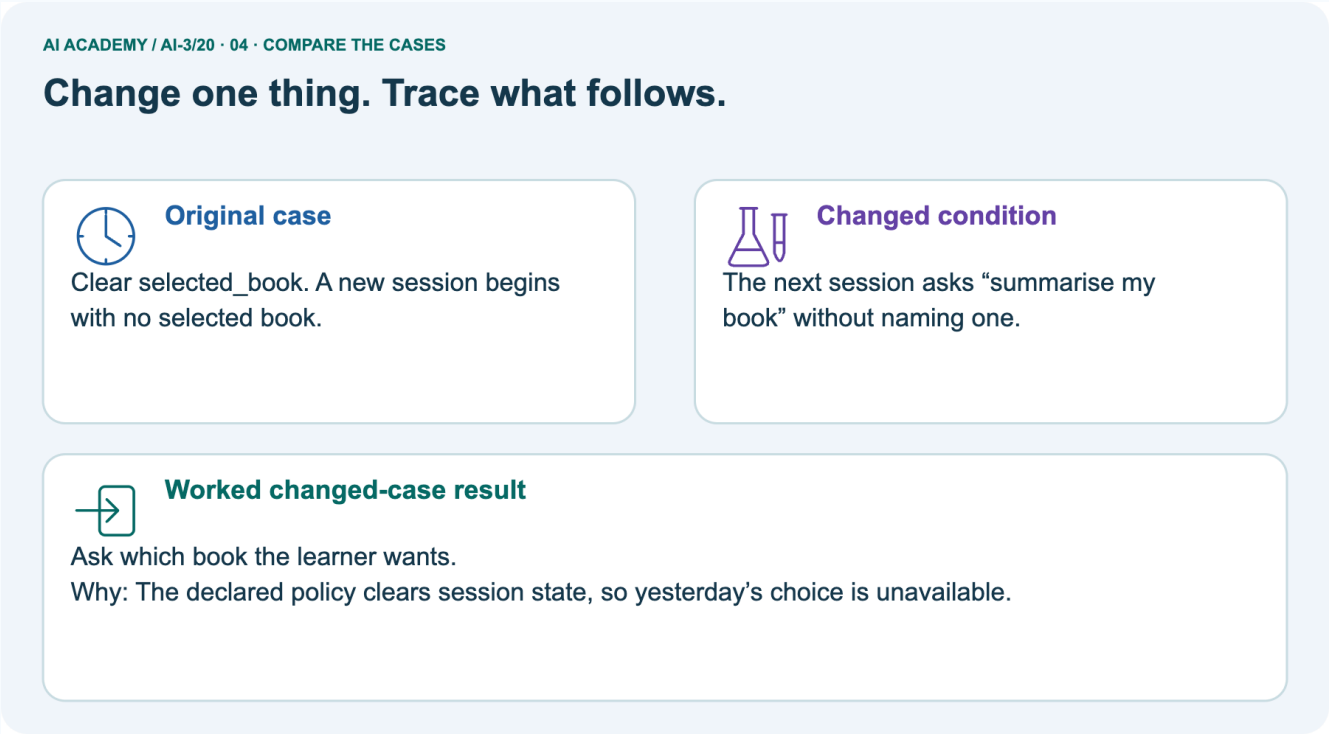
Follow the labelled picture with the matching explanation. Open the vector diagram

The selected_book **session state** is Cedar after the first choice, Maple after correction, and unset after exit. This is an application record, not proof of permanent learning.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

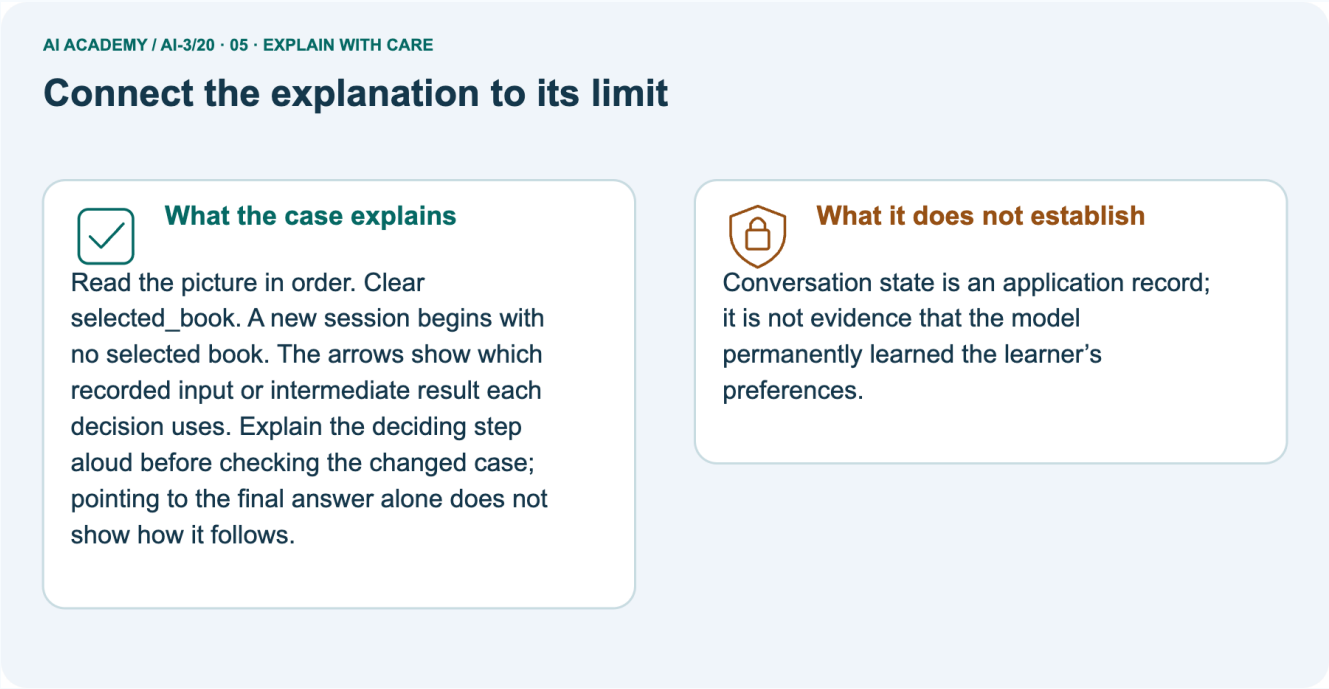
Original case: Clear selected_book. A new session begins with no selected book. Changed condition: The next session asks “summarise my book” without naming one. Expected result: Ask which book the learner wants.

Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

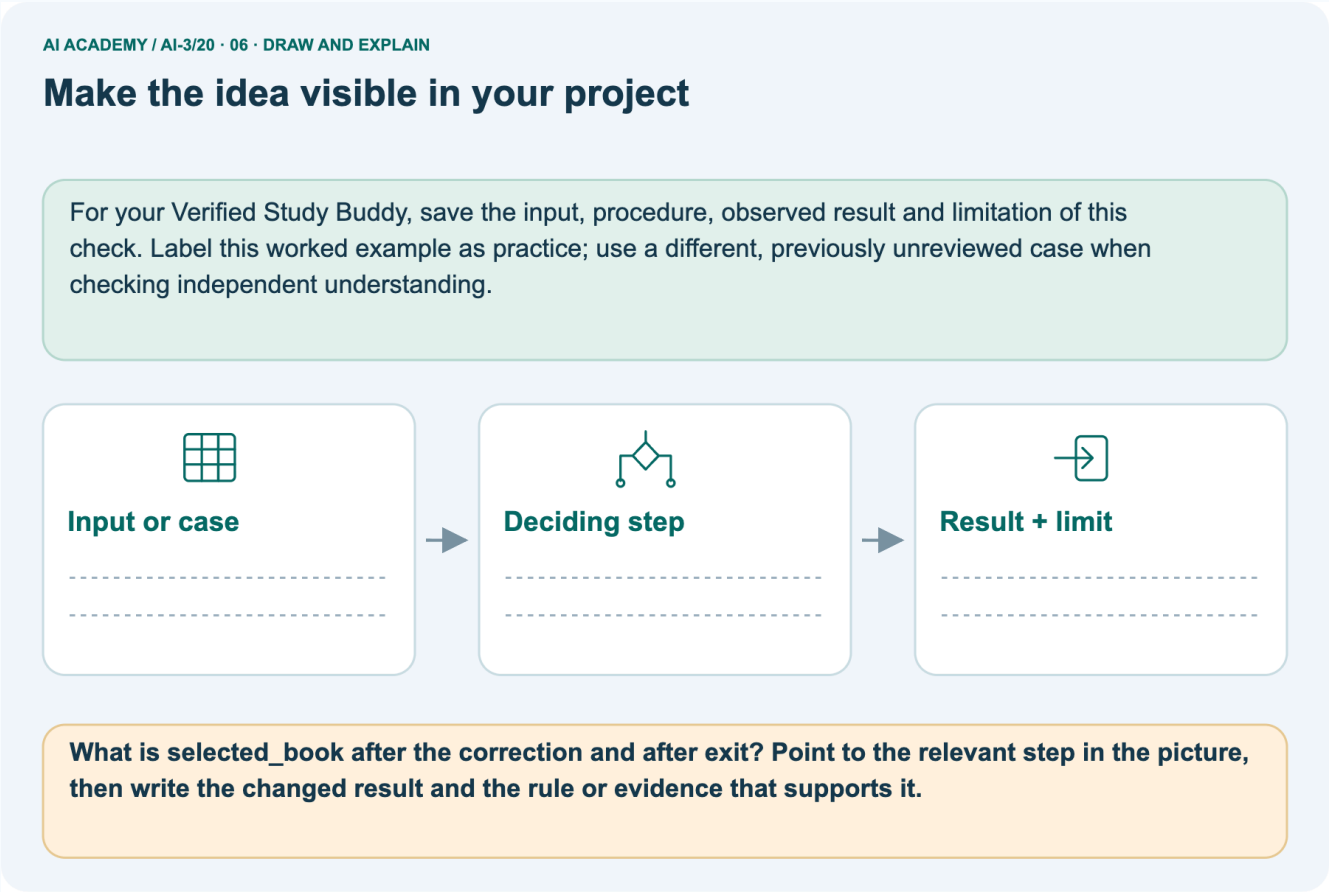
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Conversation **state** is an application record; it is not **evidence** that the model permanently learned the learner's preferences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.